

ONLINE MASTERS IN DATA SCIENCE

DSC 257R - UNSUPERVISED LEARNING

SAMPLING BY RANDOM WALK

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The Sampling Methods We'll Cover

- 1 Gibbs sampler
- 2 Langevin sampler

High-level approach: **random walk**.

Sampling by Random Walk

Want to sample from a distribution P over some space $\mathcal{X}$. This might be

- **discrete**, e.g., $\mathcal{X} = \{0,1\}^N$ (say, binary images on N pixels), or
- **continuous**, e.g., $\mathcal{X} = \mathbb{R}^d$ (say, rainfall levels in d cities)

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- $\mathcal{X}$ might be huge or infinite: we cannot enumerate all outcomes.
- We might not be able to evaluate $P(x)$ explicitly for $x \in \mathcal{X}$ due to unknown normalization factor. But can often get ratios $P(x)/P(x')$.

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Solution strategy: **do a random walk on $\mathcal{X}$**

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Want to sample from a distribution P over some space $\mathcal{X}$.

- Start at any $x \in \mathcal{X}$
- Repeatedly move to a “nearby” state, with some transition probabilities
- After a while: the distribution over the current location is (close to) P